Physics I

Code : EBA1110

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Course Content

- Lecture 1: Units and Vectors.
- Lecture 2: Newton Laws of motion &Gravitational Force, weight and mass
- Lecture 3: Work, Kinetic and Potential Energy, Work Energy theorem& Conservation law of Energy

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- Lecture 4: Oscillatory motion , simple and Harmonic Motion, Introduction to mechanical waves properties , and Travelling wave
- Lecture 5: Sound wave Power, Intensity, speed in solid , liquid or gas , Sound level calculation
- Lecture 6: Superposition of waves.

Course Content

- Lecture 7: capacitors.
- Lecture 8: Calculation of the Electric Current, Resistance, Ohm's law, Power Calculation.
- Lecture 9: Electromotive force , Internal Resistance , Resistance in Series and Parallel, Kirchhoff law and RC Circuit
- Lecture 10: Magnetic Force
- Lecture 11: Biot Savart Law, Ampere Law and their applications. Magnetic force for two parallel conductors.Explanation of Faraday Law of Induction

LECTURE 5

- **Sound wave Power, Intensity, speed in solid , liquid or gas , Sound level calculation**

Sound Waves



Sound waves categories:

(1) Audible waves

lie within the range of sensitivity of the human ear. They can be generated in a variety of ways, such as by musical instruments, human voices, or loudspeakers.

(2) Infrasonic waves

have frequencies below the audible range. Elephants can use infrasonic waves to communicate with one another, even when separated by many kilometers.

(3) Ultrasonic waves

have frequencies above the audible range. You may have used a "silent" whistle to retrieve your dog. Dogs easily hear the ultrasonic sound this whistle emits, although humans cannot detect it at all. Ultrasonic waves are also used in medical imaging

Pressure Variations in Sound Waves

We describe pictorially the motion of a one dimensional longitudinal sound pulse moving through a long tube containing a compressible gas as shown in Figure 17.1. A piston at the left end can be quickly moved to the right to compress the gas and create the pulse. Before the piston is moved, the gas is undisturbed and of uniform density as represented by the uniformly shaded region in Figure 17.1a. When the piston is pushed to the right (Fig. 17.1b), the gas just in front of it is compressed (as represented by the more heavily shaded region); the pressure and density in this region are now higher than they were before the piston moved. When the piston comes to rest (Fig. 17.1c), the compressed region of the gas continues to move to the right, corresponding to a longitudinal pulse traveling through the tube with speed v.

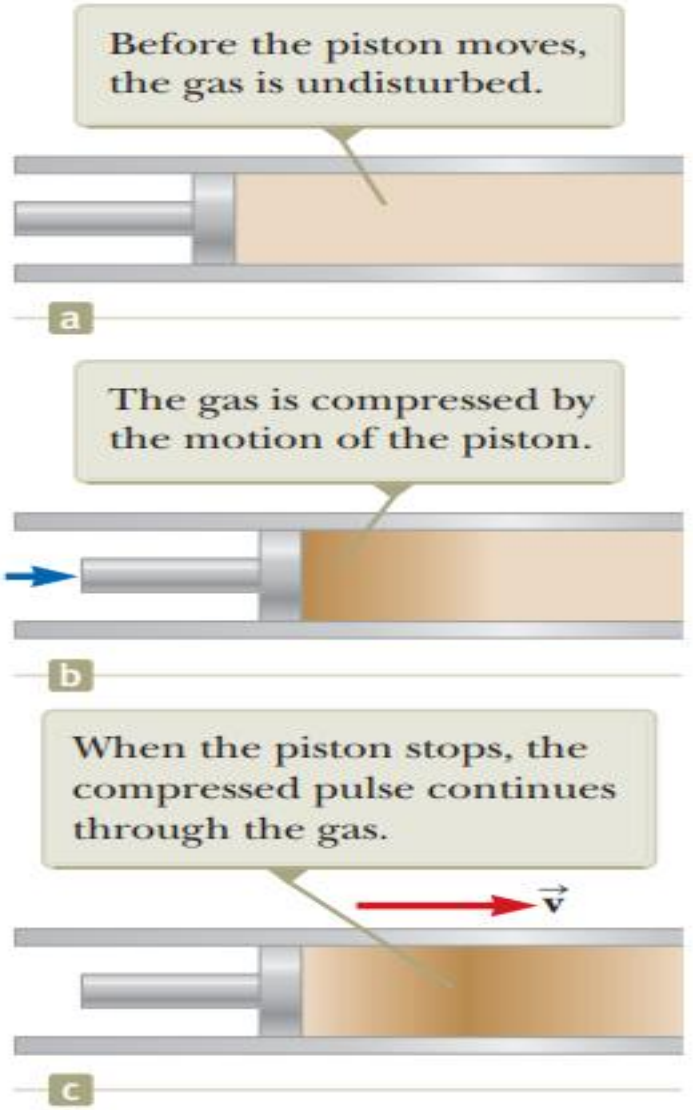


Figure 17.1 Motion of a longitudinal pulse through a compressible gas. The compression (darker region) is produced by the moving piston.

One can produce a one-dimensional periodic sound wave in the tube of gas in Figure 17.1 by causing the piston to move in simple harmonic motion. as shown in Figure 17.2. The darker parts of the colored areas in this figure represent regions in which the gas is compressed and the density and pressure are above their equilibrium values. A compressed region is formed whenever the piston is pushed into the tube.

This compressed region, called **a compression**, moves through the tube, continuously compressing the region just in front of itself.

When the piston is pulled back, the gas in front of it expands and the pressure and density

in this region fall below their equilibrium values (represented by the lighter parts of the colored areas in Fig. 17.2).

These low-pressure regions, called **rarefactions**, also propagate along the tube, following the compressions. Both regions move at the speed of sound in the medium.

As the piston oscillates sinusoidally, regions of compression and rarefaction are continuously set up.

The distance between two successive compressions (or two successive rarefactions) equals the **wavelength** λ of the sound wave.

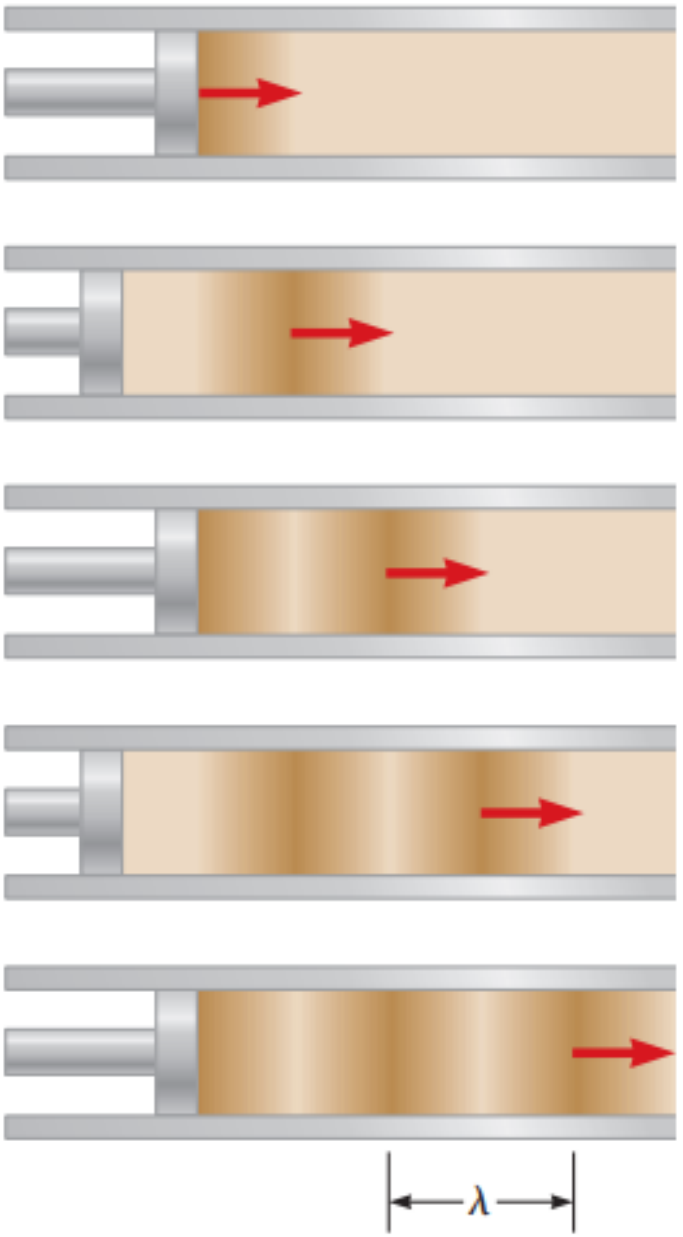


Figure 17.2 A longitudinal wave propagating through a gas-filled tube. The source of the wave is an oscillating piston at the left.

If s(x, t) is the position of a small element relative to its equilibrium position, we can express this harmonic position function as

$$s(x, t) = s_{\max} \cos (kx - \omega t) \tag{17.1}$$

where $s_{\max}$ is the maximum position of the element relative to equilibrium. This parameter is often called the **displacement amplitude** of the wave. The parameter k is the wave number, and ω is the angular frequency of the wave. Notice that the displacement of the element is along x, in the direction of propagation of the sound wave.

The variation in the gas pressure ΔP measured from the equilibrium value is also periodic with the same wave number and angular frequency as for the displacement in Equation 17.1. Therefore, we can write

$$\Delta P = \Delta P_{\max} \sin (kx - \omega t) \tag{17.2}$$

where **the pressure amplitude** $\Delta P_{\max}$ is the maximum change in pressure from the equilibrium value.

Consider the piston–tube arrangement of Figure 17.1 once again. In Figure 17.3a, we focus our attention on a small cylindrical element of undisturbed gas of length Δx and area A. The volume of this element is $V_i = A\,\Delta x$.

Figure 17.3b shows this element of gas after a sound wave has moved it to a new position. The cylinder's two flat faces move through different distances s_1 and s_2. The change in volume ΔV of the element in the new position is equal to $A\,\Delta s$, where $\Delta s = s_1 - s_2$.

From the definition of bulk modulus (see Eq. 12.8), we express the pressure variation in the element of gas as a function of its change in volume:

$$\Delta P = -B\frac{\Delta V}{V_i}$$

Let's substitute for the initial volume and the change in volume of the element:

$$\Delta P = -B\frac{A\,\Delta s}{A\,\Delta x}$$

Let the length Δx of the cylinder approach zero so that the ratio $\Delta s/\Delta x$ becomes a partial derivative:

$$\Delta P = -B\frac{\partial s}{\partial x} \qquad (17.3)$$

Substitute the position function given by Equation 17.1:

$$\Delta P = -B\frac{\partial}{\partial x}[s_{\max}\cos(kx - \omega t)] = Bs_{\max}k\,\sin(kx - \omega t)$$

We also see that the displacement and pressure amplitudes are related by

$$\Delta P_{\max} = Bs_{\max}k \qquad (17.4)$$

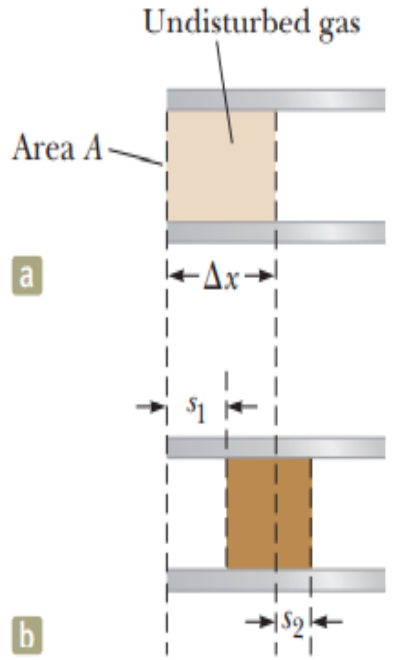


Figure 17.3 (a) An undisturbed element of gas of length Δx in a tube of cross-sectional area A. (b) When a sound wave propagates through the gas, the element is moved to a new position and has a different length. The parameters s_1 and s_2 describe the displacements of the ends of the element from their equilibrium positions.

. A comparison of Equations 17.1 and 17.2 shows that the pressure wave is 90 ° out of phase with the displacement wave.

Graphs of these functions are shown in Figure 17.4.

The pressure variation is a maximum when the displacement from equilibrium is zero, and the displacement from equilibrium is a maximum when the pressure variation is zero.

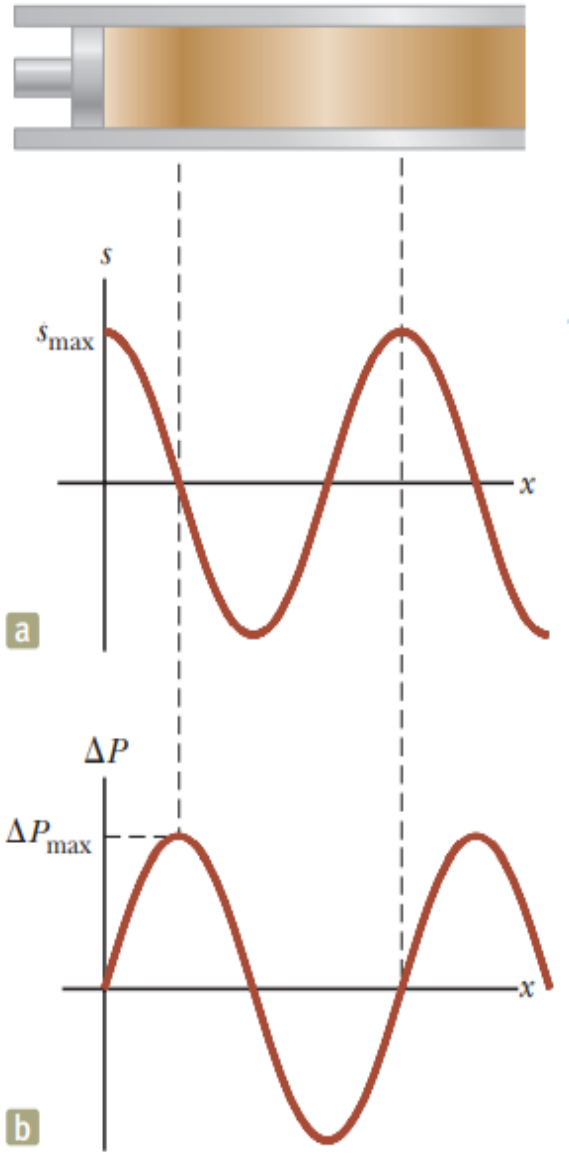


Figure 17.4 (a) Displacement amplitude and (b) pressure amplitude versus position for a sinusoidal longitudinal wave.

Speed of Sound Waves

the speed of sound in a gas:

$$v = \sqrt{\frac{B}{\rho}} \qquad (17.8)$$

It is interesting to compare this expression with the speed of transverse waves on a string,

$$v = \sqrt{T/\mu}.$$

In both cases, the wave speed depends on an elastic property of the medium (bulk modulus B or string tension T) and on an inertial property of the medium (volume density ρ or linear density μ). In fact, the speed of all mechanical waves follows an expression of the general form

$$v = \sqrt{\frac{\text{elastic property}}{\text{inertial property}}}$$

For longitudinal sound waves in a solid rod of material, for example, the speed of sound depends on Young's modulus Y and the density ρ. Table 17.1 (page 512) provides the speed of sound in several different materials.

The speed of sound also depends on the temperature of the medium. For sound traveling through air, the relationship between wave speed and air temperature is

$$v = 331\sqrt{1 + \frac{T_C}{273}} \qquad (17.9)$$

Table 17.1 Speed of Sound in Various Media

Medium	v (m/s)	Medium	v (m/s)	Medium	v (m/s)
Gases		**Liquids at 25°C**		**Solids**[a]	
Hydrogen (0°C)	1 286	Glycerol	1 904	Pyrex glass	5 640
Helium (0°C)	972	Seawater	1 533	Iron	5 950
Air (20°C)	343	Water	1 493	Aluminum	6 420
Air (0°C)	331	Mercury	1 450	Brass	4 700
Oxygen (0°C)	317	Kerosene	1 324	Copper	5 010
		Methyl alcohol	1 143	Gold	3 240
		Carbon tetrachloride	926	Lucite	2 680
				Lead	1 960
				Rubber	1 600

Having an expression (Eq. 17.8) for the speed of sound, we can now express the relationship between pressure amplitude and displacement amplitude for a sound wave (Eq. 17.4) as

$$\Delta P_{\max} = B s_{\max} k = (\rho v^2) s_{\max} \left(\frac{\omega}{v}\right) = \rho v \omega s_{\max} \tag{17.10}$$

This expression is a bit more useful than Equation 17.4 because the density of a gas is more readily available than is the bulk modulus.

Power and Intensity of Periodic Sound Waves

Naturally, we would expect sound waves to also represent a transfer of energy. Consider the element of gas acted on by the piston in Figure 17.5. Imagine that the piston is moving back and forth in simple harmonic motion at angular frequency ω. Imagine also that the length of the element becomes very small so that the entire element moves with the same velocity as the piston. Then we can model the element as a particle on which the piston is doing work.

The rate at which the piston is doing work on the element at any instant of time is given by Equation 8.19:

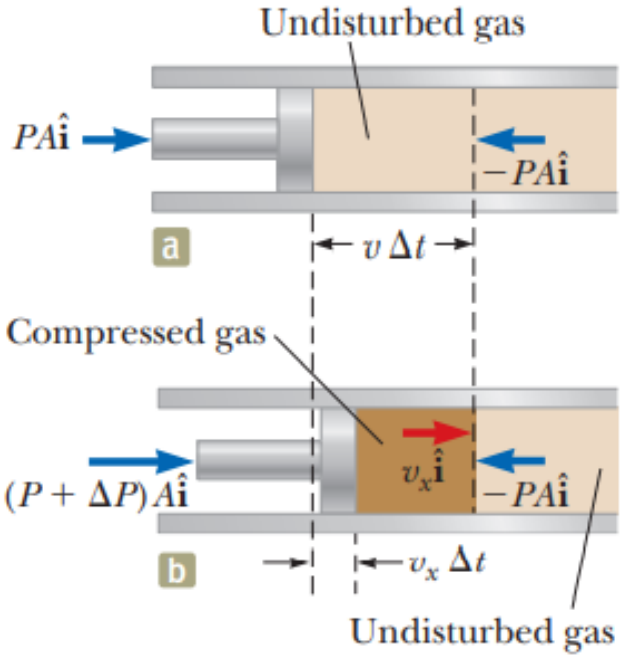


Figure 17.5 (a) An undisturbed element of gas of length $v\ \Delta t$ in a tube of cross-sectional area A. The element is in equilibrium between forces on either end. (b) When the piston moves inward at constant velocity v_x due to an increased force on the left, the element also moves with the same velocity.

$$Power = \vec{\mathbf{F}} \cdot \vec{\mathbf{v}}_x$$

where we have used *Power* rather than P so that we don't confuse power P with pressure P! The force $\vec{\mathbf{F}}$ on the element of gas is related to the pressure and the velocity $\vec{\mathbf{v}}_x$ of the element is the derivative of the displacement function, so we find

$$Power = [\Delta P(x, t)A]\hat{\mathbf{i}} \cdot \frac{\partial}{\partial t}[s(x, t)\hat{\mathbf{i}}]$$

$$= [\rho v \omega A s_{\max} \sin(kx - \omega t)]\left\{\frac{\partial}{\partial t}[s_{\max} \cos(kx - \omega t)]\right\}$$

$$= \rho v \omega A s_{\max} \sin(kx - \omega t)][\omega s_{\max} \sin(kx - \omega t)]$$

$$= \rho v \omega^2 A s_{\max}^2 \sin^2(kx - \omega t)$$

We now find the time average power over one period of the oscillation. For any given value of x, which we can choose to be $x = 0$, the average value of $\sin^2(kx - \omega t)$ over one period T is

$$\frac{1}{T}\int_0^T \sin^2(0 - \omega t)\, dt = \frac{1}{T}\int_0^T \sin^2 \omega t\, dt = \frac{1}{T}\left(\frac{t}{2} + \frac{\sin 2\omega t}{2\omega}\right)\Bigg|_0^T = \tfrac{1}{2}$$

therefore

$$(\text{Power})_{\text{avg}} = \tfrac{1}{2}\rho v \omega^2 A S_{MAX}^2$$

the intensity *I* of a wave

We define the intensity I of a wave, or the power per unit area, as the rate at which the energy transported by the wave transfers through a unit area A perpendicular to the direction of travel of the wave

$$I \equiv \frac{(Power)_{avg}}{A}$$

In this case, the intensity is therefore

$$I = \tfrac{1}{2}\rho v(\omega s_{max})^2$$

This expression can also be written in terms of the pressure amplitude ΔP_{max}; in this case, we use Equation 17.10 to obtain

$$I = \frac{(\Delta P_{max})^2}{2\rho v} \qquad (17.12)$$

Consider the special case of a point source emitting sound waves equally in all directions.

If the air around the source is perfectly uniform, the sound power radiated in all directions is the same, and the speed of sound in all directions is the same. The result in this situation is called a spherical wave.

Figure 17.6 shows these spherical waves as a series of circular arcs concentric with the source. Each arc represents a surface over which the phase of the wave is constant. We call such a surface of constant phase a wave front.

The radial distance between adjacent wave fronts that have the same phase is the wavelength λ of the wave.

The radial lines pointing outward from the source, representing the direction of propagation of the waves, are called rays.

The average power emitted by the source must be distributed uniformly over each spherical wave front of area $4\pi r^2$. Hence, the wave intensity at a distance r from the source is

$$I = \frac{(Power)_{avg}}{A} = \frac{(Power)_{avg}}{4\pi r^2}$$

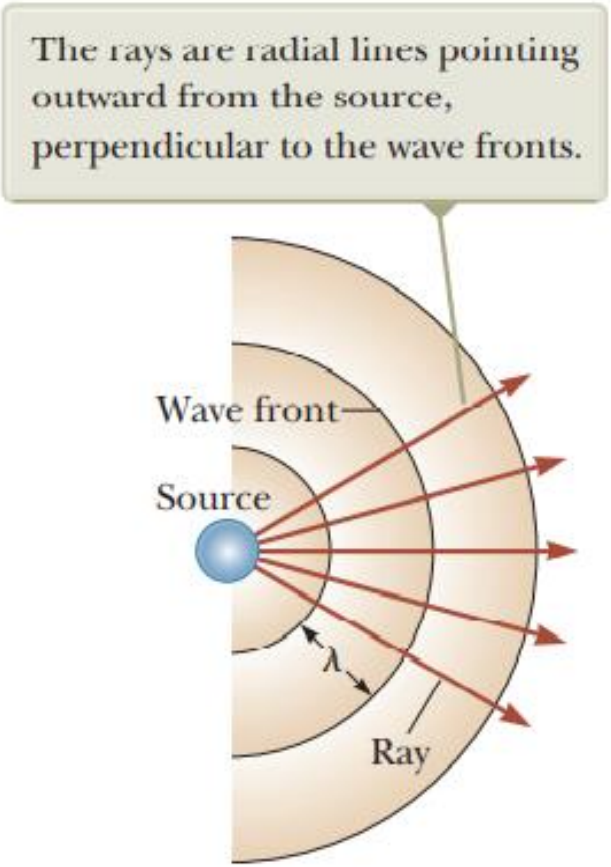


Figure 17.6 Spherical waves emitted by a point source. The circular arcs represent the spherical wave fronts that are concentric with the source.

Sound Level in Decibels

sound level b (Greek letter beta) is defined by the equation:

$$\beta \equiv 10 \log\left(\frac{I}{I_0}\right)$$

The constant I_0 is the *reference intensity*, taken to be at the threshold of hearing ($I_0 = 1.00 \times 10^{-12}$ W/m^2), and I is the intensity in watts per square meter to which the sound level β corresponds, where β is measured[2] in **decibels** (dB). On this scale, the threshold of pain ($I = 1.00$ W/m^2) corresponds to a sound level of $\beta = 10 \log [(1 \text{ W/m}^2)/(10^{-12} \text{ W/m}^2)] = 10 \log (10^{12}) = 120$ dB, and the threshold of hearing corresponds to $\beta = 10 \log [(10^{-12} \text{ W/m}^2)/(10^{-12} \text{ W/m}^2)] = 0$ dB.

Prolonged exposure to high sound levels may seriously damage the human ear. Ear plugs are recommended whenever sound levels exceed 90 dB. Recent evidence suggests that "noise pollution" may be a contributing factor to high blood pressure, anxiety, and nervousness. Table 17.2 gives some typical sound levels.

Table 17.2

Sound Levels

Source of Sound	β (dB)
Nearby jet airplane	150
Jackhammer; machine gun	130
Siren; rock concert	120
Subway; power lawn mower	100
Busy traffic	80
Vacuum cleaner	70
Normal conversation	60
Mosquito buzzing	40
Whisper	30
Rustling leaves	10
Threshold of hearing	0

Example 17.3 Sound Levels

Two identical machines are positioned the same distance from a worker. The intensity of sound delivered by each operating machine at the worker's location is $2.0 \times 10^{-7}\ \mathrm{W/m^2}$.

(A) Find the sound level heard by the worker when one machine is operating.

SOLUTION

Conceptualize Imagine a situation in which one source of sound is active and is then joined by a second identical source, such as one person speaking and then a second person speaking at the same time or one musical instrument playing and then being joined by a second instrument.

Categorize This example is a relatively simple analysis problem requiring Equation 17.14.

Analyze Use Equation 17.14 to calculate the sound level at the worker's location with one machine operating:

$$\beta_1 = 10 \log\left(\frac{2.0 \times 10^{-7}\ \mathrm{W/m^2}}{1.00 \times 10^{-12}\ \mathrm{W/m^2}}\right) = 10 \log\left(2.0 \times 10^{5}\right) = 53\ \mathrm{dB}$$

(B) Find the sound level heard by the worker when two machines are operating.

SOLUTION

Use Equation 17.14 to calculate the sound level at the worker's location with double the intensity:

$$\beta_2 = 10 \log\left(\frac{4.0 \times 10^{-7}\ \mathrm{W/m^2}}{1.00 \times 10^{-12}\ \mathrm{W/m^2}}\right) = 10 \log\left(4.0 \times 10^{5}\right) = 56\ \mathrm{dB}$$